

```

#include<iostream.h>
#include<fstream.h>
#include<iomanip.h>
#include<math.h>
#include<conio.h>
ofstream output("rl.txt");
float f(float I,float R,float L,float V)
{
return ((V-(I*R))/L);
}
class RL
{
int i,n;
float t,h,R,L,V,V2,I,I0,O,o,W,w,Ti,Ti1,Ti2,Tf,K1,K2,K3,K4,Pr,Pv,Pl;
public:
void getdata();
void growth();
void decay();
void alternating();
};
void RL::getdata()
{
cout<<"Enter the value of resistance"<<endl;
cin>>R;
cout<<"Enter the value of Inductance"<<endl;
cin>>L;
cout<<"Enter the value of initial time"<<endl;
cin>>Ti;
cout<<"Enter the value of final time"<<endl;
cin>>Tf;
cout<<"Enter the value of time step"<<endl;
cin>>h;
}
void RL::growth()
{
V=1.0;
I=0.0;
Ti1=0.0;
Pr=0.0;
Pv=0.0;
Pl=0.0;
n=(Tf-Ti)/h;
output<<"Growth of current"<<endl;
output<<"Time"<<setw(20)<<"Current"<<setw(20)<<"Pv"<<setw(15)<<"Pr"<<setw(15)<<"Pl"
<<endl;
for(i=0;i<=n;i++)
{
K1=h*f(I,R,L,V);
K2=h*f(I+(K1/2),R,L,V);
K3=h*f(I+(K2/2),R,L,V);
K4=h*f(I+K3,R,L,V);
I=I+(K1+2*K2+2*K3+K4)/6;
Pr=I*I*R;
Pv=V*I;
Pl=L*I*f(I,R,L,V);
Ti1=Ti1+h;
output<<Ti1<<setw(20)<<I<<setw(20)<<Pr<<setw(15)<<Pv<<setw(15)<<Pl<<endl;
}
}

```

```

void RL::decay()
{
V=0.0;
I=1.0;
Ti1=0.0;
n=(Tf-Ti)/h;
output<<"DECAY OF CURRENT"<<endl;
output<<"Time"<<setw(20)<<"Current"<<endl;
for(i=0;i<=n;i++)
{
K1=h*f(I,R,L,V);
K2=h*f(I+(K1/2),R,L,V);
K3=h*f(I+(K2/2),R,L,V);
K4=h*f(I+K3,R,L,V);
I=I+(K1+2*K2+2*K3+K4)/6;
Ti1=Ti1+h;
output<<Ti1<<setw(20)<<I<<endl;
}
}
void RL::alternating()
{
V2=2.0;
Ti2=0.0;
w=1.0;
n=(Tf-Ti)/h;
output<<"ALTERNATING CURRENT"<<endl;
output<<"Time"<<setw(20)<<"Current"<<endl;
W=2*3.142*w;
O=(atan((-W*L)/R))*(3.142/180);
I0=V2/(sqrt((R*R)+(W*W*L*L)));
for(i=0;i<=n;i++)
{
I=I0*cos(W*Ti2+O);
output<<Ti2<<setw(20)<<I<<endl;
Ti2=Ti2+h;
}
}
int main()
{
RL p;
p.getdata();
p.growth();
p.decay();
p.alternating();
return(0);
}

```